

Structure and Properties of Mirror-Twin Boundary Networks in Substoichiometric MoTe_2 : Insights from First-Principles Calculations

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Defect engineering in two-dimensional (2D) materials offers a powerful route to emergent electronic phases beyond those accessible in pristine crystals. The experiments [1] demonstrate that in substoichiometric MoTe_2 , point defects condense into extended mirror-twin boundaries (MTBs) that self-organize into periodic networks, forming a homologous series of novel crystalline phases. Scanning tunneling microscopy reveals that these phases consist of periodically arranged triangular MTB loops, while our density functional theory calculations show that the experimentally observed structures lie on or near the convex hull, providing a unified thermodynamic framework for their stability. The preferred structural motif arises from energetically favorable vertex configurations and specific relative offsets between MTB domains. Electronic structure calculations further demonstrate that these ordered MTB networks give rise to Kagome-like flat bands with pronounced Van Hove singularities, and in certain cases, emergent ferromagnetism. These findings establish MTB phases not as mere defect arrangements but as distinct 2D crystals with tunable quantum properties. Our results highlight a general strategy for engineering correlated electronic states through defect ordering in low-dimensional materials.

[1] Onyedikachi Alanwoko, Daniil Kruklinkii, Mahdi Ghorbani-Asl, Arkady V. Krashennnikov, Matthias Batzill, *Nano Letters* **26**, 376–383 (2026).